

ComPDF Conversion SDK for C++ 4.1.0

Upgrade Warnings

This document contains a list of considerations that developers should be aware of when upgrading from previous versions to ComPDF Conversion SDK for C++ 4.1.0. Changes in the code may be necessary to ensure compatibility with the new version for your project to compile properly.

Upgrading from version 4.0 to version 4.1

If you are upgrading from version 4.0 (4.0.0 / 4.0.1) to version 4.1.0, please note the following changes between 4.0 and 4.1. Several of them are **source-incompatible** and require code updates before recompiling.

Breaking changes in the `ConvertOptions` struct (ABI hardening):

To eliminate cross-ABI risk introduced by STL containers crossing the SDK boundary, `ConvertOptions` no longer uses `std::string` or `std::vector`. All variable-length fields are now C-style fixed-size arrays:

Field	4.0 type	4.1 type
<code>font_name</code>	<code>std::string</code>	<code>char font_name[kConvertOptionStringBufferSize]</code> (256 bytes)
<code>page_ranges</code>	<code>std::string</code>	<code>char page_ranges[kConvertOptionStringBufferSize]</code> (256 bytes)
<code>languages</code>	<code>std::vector<OCRLanguage></code>	<code>OCRLanguage languages[kConvertOptionMaxLanguages]</code> (max 32)
<code>language_count</code>	<i>(did not exist)</i>	<code>int language_count = 0</code> — number of valid entries in <code>languages</code>

Migration:

```
// 4.0
ConvertOptions opt;
opt.font_name    = "Arial";
opt.page_ranges  = "1-3,5";
opt.languages    = { OCRLanguage::e_English, OCRLanguage::e_Chinese };

// 4.1
ConvertOptions opt;
std::snprintf(opt.font_name,    sizeof(opt.font_name),    "%s", "Arial");
std::snprintf(opt.page_ranges,  sizeof(opt.page_ranges),  "%s", "1-3,5");
opt.languages[0]    = OCRLanguage::e_English;
opt.languages[1]    = OCRLanguage::e_Chinese;
opt.language_count  = 2;
```

The `<string>` and `<vector>` includes have been removed from `base_type.h`; if your code relied on transitive inclusion, add the headers explicitly.

New field in the `ConvertOptions` struct:

- `bool enable_ai_table_recognition = true` — independently controls the AI table recognition stage. It is enabled by default and only takes effect when layout analysis is on (`enable_ai_layout = true` or implicitly via `enable_ocr = true`). Setting it to `false` disables the table model even when layout analysis is active.

Extended scope of the `font_name` option:

In 4.0, `font_name` was honored only for PDF → Word / Excel / PowerPoint. In 4.1 it additionally applies to:

- PDF → Searchable PDF
- PDF → OFD

For Searchable PDF and OFD, `font_name` controls the font used for the text layer overlaid on the page background.

Breaking changes in `CConvertCallback` (custom AI integration):

The 4.0 single-method OCR callback has been replaced by a trigger / result-getter pattern that is shared by OCR, Layout Analysis, and Table Recognition.

`COCRCallback` signature change:

```
// 4.0
typedef bool (*COCRCallback)(const char* image_path, bool enable_ocr);

// 4.1
typedef bool (*COCRCallback)(const char* image_path);
```

New typedefs added in `com_config_macros.h`:

- `CLayoutCallback`, `CTableCallback` — trigger callbacks for layout analysis and table recognition.
- `CGetOCRResultCallback`, `CGetLayoutResultCallback`, `CGetTableResultCallback` — paired getter callbacks that return the inference result as a UTF-8 JSON C-string.

New fields appended to `CConvertCallback` (all default to `nullptr`):

```
COCRCallback          ocr;
CLayoutCallback       layout;
CTableCallback        table;
CGetOCRResultCallback get_ocr_result;
CGetLayoutResultCallback get_layout_result;
CGetTableResultCallback get_table_result;
```

When a trigger / getter pair is registered, the SDK skips its built-in DocumentAI invocation for that capability and consumes your JSON output instead. If a pair is left as `nullptr`, the SDK falls back to the built-in model. See the chapter *Use Custom AI Models via Callbacks* in the Developer Guide for the JSON schema and a full integration example.

Removed API:

- `CPDFConversion::Cancel()` — global cancellation has been removed. Use per-task cancellation by returning `true` from the `cancel` callback in `CConvertCallback`, which the SDK polls between pages.
- The corresponding C API `CPDF_Cancel()` has also been removed.

New API in `LibraryManager`:

- `static int32_t GetRemainingPageQuota()` — returns the remaining page quota associated with the current license.
 - `>= 0`: actual remaining pages.
 - `-1`: not initialized.
 - `-2`: unlimited.
 - `-3`: quota file corrupted.

New entries in the `ErrorCode` enum class:

- `e_ErrLicenseConcurrencyExceeded = 30` — the current conversion concurrency exceeds the license limit.
- `e_ErrLicensePageLimitExceeded = 31` — the current conversion pages exceed the license limit.
- `e_ErrLicenseQuotaCorrupted = 32` — the quota persistence file has been tampered with.

Existing error codes are unchanged. If you have a `switch` over `ErrorCode`, consider adding handling for the new license-related codes.